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Synthetic Methods

Highly Selective Thiiranation of 1,2-Allenyl Sulfones with Br_2 and $Na_2S_2O_3$: Mechanism and Asymmetric Synthesis of Alkylidenethiiranes**

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Electrophilic addition reactions of allenes have been explored in the search for highly selective reactions for organic synthesis.[1] We demonstrated the electrophilic halocyclization of allenes with a nucleophilic functionality.[2] On the basis of these observations, we developed highly regio- and stereoselective halohydroxylation reactions of heteroatom-substituted allenes, including 1,2-allenyl sulfides,[3] selenides,[4] and sulfoxides.[5] In these reactions, the stereoselectivity depends largely on the nature of the heteroatom: The reactions of sulfides or selenides afforded *Z* products, whereas sulfoxides reacted to give *E* products. However, under the reaction conditions used for the halohydroxylation of 1,2-allenyl sulfoxides,[5] no reaction was observed for 1,2-allenyl sulfones, probably as a result of the strong electron-withdrawing ability of the sulfone group.[6] Padwa and co-workers reported the electrophilic addition of Br_2 and I_2 to 1,2-propadienyl phenyl sulfone to give the corresponding *E*-2,3-dihalopropenyl phenyl sulfone with high stereoselectivity.[7] Furthermore, Braverman and co-workers reported the electrophilic addition of Br_2 to bisallenyl sulfones to form γ sultines.[8] Herein we disclose our recent observations on the electrophilic reaction of 1,2-allenyl sulfones with Br_2 and $Na_2S_2O_3$ to afford alkylidenethiiranes[9] with high regio- and stereoselectivity.

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Supporting information for this article (including a typical experimental procedure and analytical data for all products not listed in the text) is available on the WWW under http://www.angewandte.org or from the author.

We initiated our study by attempting the reaction of 1,2-butadienyl phenyl sulfone (**1a**) with Br_2. We observed with interest that instead of the expected halohydroxylation product (*E*)-**2a**,[3–5] the reaction of **1a** with Br_2 (1.5 equiv) in MeCN in the presence of H_2O (2.0 equiv) afforded the 1-(phenylsulfonyl)methylene-2-methylthiirane (*E*)-**3a** in 56% yield together with the 2,3-dibromobutenyl sulfone (*E*)-**4a** (11%)[7] upon sequential quenching with water and aqueous $Na_2S_2O_3$ (Table 1, entry 1). A similar result was obtained in

Table 1: Electrophilic reaction of 1,2-butadienyl phenyl sulfone (**1a**) with Br_2.



Entry	H_2O [equiv]	Br_2 [equiv]	*t* [min]	Yield of **3a** [%]	Yield of **4a** [%]
1	2.0	1.5	60	56	11
2	0	1.5	60	56	11
3	0	2.0	60	67	9
4	0	2.5	15	70	–
5[a]	0	2.5	15	67	1.5

[a] The reaction was quenched directly with aqueous $Na_2S_2O_3$.

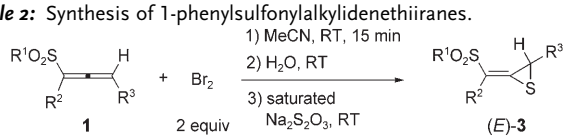
the absence of H_2O (Table 1, entry 2); with 2 equivalents of Br_2, (*E*)-**3a** and (*E*)-**4a** were formed in 67 and 9% yield, respectively (Table 1, entry 3). Finally, we observed that the reaction of **1a** with 2.5 equivalents of Br_2 in MeCN afforded (*E*)-**3a** in 70% yield as the only product (Table 1, entry 4).

We found that the reaction was quite general: When different groups R^1, R^2, and R^3 were introduced, the reaction afforded substituted 1-sulfonyl alkylidenethiiranes (*E*)-**3** in moderate to good yields (Table 2). The structure of one of these products, (*E*)-**3m**, was confirmed by X-ray diffraction analysis (see the Supporting Information).[10] When the reaction of **1b** with Br_2 was quenched directly with saturated, aqueous $Na_2S_2O_3$, the yield was much lower (Table 2, entries 2 and 3).

As optically active 1,2-allenyl sulfones can be prepared very conveniently by a two-step procedure from readily available optically active propargylic alcohols,[11,5b] we also studied the possibility of using this reaction for the preparation of optically active 1-sulfonyl alkylidenethiiranes. The axial chirality in the 1,2-allenyl sulfones was transferred with high efficiency to the alkylidenethiiranes in the form of

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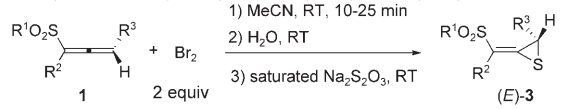
Table 2: Synthesis of 1-phenylsulfonylalkylidenethiiranes.



Entry	1	R^1	R^2	R^3	Yield of (*E*)-3 [%]
1[a,b]	**1a**	Ph	H	CH_3	70 (**3a**)
2	**1b**	Ph	H	C_2H_5	70 (**3b**)
3[c]	**1b**	Ph	H	C_2H_5	55 (**3b**)
4	**1c**	Ph	H	*n*-C_3H_7	68 (**3c**)
5	**1d**	Ph	H	*n*-C_4H_9	71 (**3d**)
6	**1e**	Ph	H	*n*-C_5H_{11}	63 (**3e**)
7	**1f**	Ph	H	*n*-C_6H_{13}	69 (**3f**)
8	**1g**	Ph	H	Bn	72 (**3g**)
9	**1h**	Ph	*n*-C_4H_9	CH_3	80 (**3h**)
10	**1i**	Ph	*n*-C_4H_9	C_2H_5	67 (**3i**)
11	**1j**	Ph	*n*-C_4H_9	Bn	84 (**3j**)
12[d]	**1k**	Ph	C_2H_5	CH_3	84 (**3k**)
13[e]	**1l**	*p*-BrC_6H_4	C_2H_5	CH_3	61 (**3l**)
14[e]	**1m**	*p*-BrC_6H_4	H	CH_3	65 (**3m**)

[a] In this reaction 2.5 equivalents of Br_2 were used. [b] Reaction time: 10 min. [c] The reaction mixture was quenched directly with aqueous $Na_2S_2O_3$. [d] Reaction time: 20 min. [e] Reaction time: 25 min. Bn = benzyl.

Table 3: Synthesis of optically active 1-phenylsulfonylalkylidenethiiranes.



1 (*ee* [%])[a]	R^1	R^2	R^3	Yield of (*E*)-3 [%]	*ee* [%][a]
(*R*)-**1f** (>99)	Ph	H	*n*-C_6H_{13}	58 ((*S*)-(*E*)-**3f**)	99
(*R*)-**1e** (99)	Ph	H	*n*-C_5H_{11}	62 ((*S*)-(*E*)-**3e**)	99
(*R*)-**1l** (96)	*p*-BrC_6H_4	C_2H_5	CH_3	74 ((*S*)-(*E*)-**3l**)	96
(*R*)-**1m** (99)	*p*-BrC_6H_4	H	CH_3	57 ((*S*)-(*E*)-**3m**)	99

[a] The *ee* values were determined by HPLC on a chiral phase (see the Supporting Information for details).

central chirality (Table 3). The absolute configuration of the products was determined by X-ray diffraction analysis of (*S*)-(*E*)-**3m**[12] by using the bromine atom in the molecule as a reference (Figure 1).

To study the mechanism of the reaction and the origin of the regio- and stereoselectivity, 1,2-butadienyl *p*-bromophenyl sulfone (**1l**) was prepared. The treatment of **1l** with

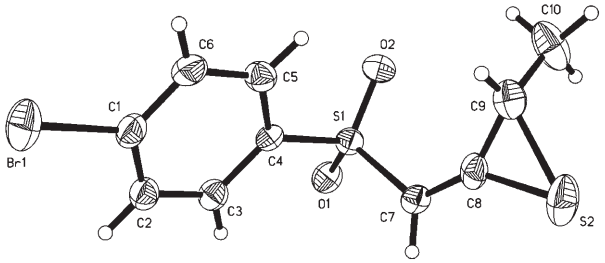


Figure 1. ORTEP representation of (*S*)-(*E*)-**3m**.

Br_2 (2.0 equiv) under the standard reaction conditions afforded the expected product (*E*)-**3l** in 61% yield (Table 2, entry 13). However, a five-membered intermediate **5l** with Br_3^- as the counter ion could be isolated with careful handling after the treatment of **1l** with Br_2 (2.0 equiv). The structure of this intermediate was established by X-ray diffraction analysis of its *trans* isomer (Figure 2).[13] The treatment of **5l** (*cis*/*trans* 1:1) with H_2O in MeCN followed by the addition of saturated aqueous $Na_2S_2O_3$ afforded cleanly (*E*)-**3l** as a single product (Scheme 1).

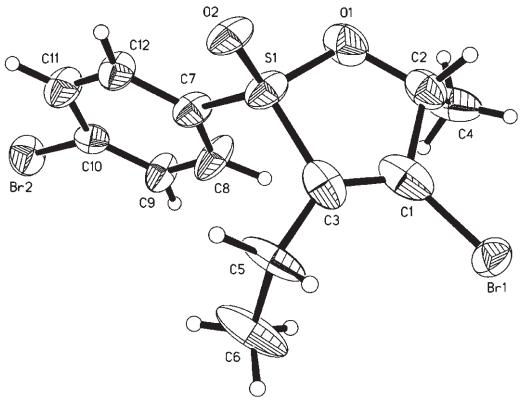
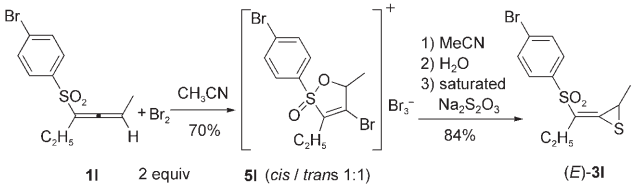


Figure 2. ORTEP representation of *trans*-**5l**.



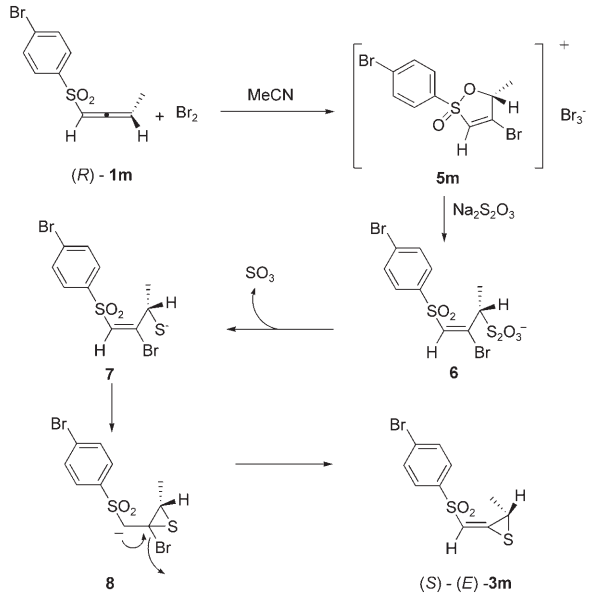
Scheme 1. Identification of the intermediate **5l** in the thiiranation of **1l** with Br_2 and $Na_2S_2O_3$.

We propose a mechanism for this transformation on the basis of the results shown in Scheme 1 and the steric outcome presented in Table 3 and Figure 1 (Scheme 2): The electrophilic addition of Br_2 to (*R*)-**1m** in MeCN may afford the 5-membered-ring intermediate **5m**, which may be attacked by $S_2O_3^{2-}$ with ring opening to afford the acyclic intermediate **6**. The release of a molecule of SO_3 would afford the 2-bromoallylic sulfide **7**, which would undergo intramolecular conjugated addition and elimination to afford (*S*)-(*E*)-**3m**. The stereoselectivity of this reaction is determined by the intermediacy of **5** and the stereospecific nature of the addition–elimination reaction of **7**.

In conclusion, we have demonstrated that the reaction of 1,2-allenyl sulfones with Br_2 followed by sequential treatment with H_2O and an aqueous solution of $Na_2S_2O_3$ affords 1-sulfonyl alkylidenethiiranes with high regio- and stereoselectivity. Axial chirality in the allenes can be converted efficiently into central chirality in the final products. This reaction should be useful in organic synthesis because

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Scheme 2. Proposed mechanism of the stereoselective thiiranation of 1,2-allenyl sulfones with Br_2 and $Na_2S_2O_3$.

optically active starting allenes, which are readily available from propargylic alcohols, are transformed into densely functionalized products.

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[11] Optically pure allenyl sulfones were synthesized by the oxidation of chiral allenyl sulfoxides with *m*-chloroperbenzoic acid in an ice bath. For the synthesis of chiral allenyl sulfoxides, see reference [5b].

[12] Crystal data for (*S*)-(*E*)-**3m**: $C_{10}H_9BrO_2S_2$, $M_r = 305.20$, monoclinic, $P2(1)/n$, $Mo_{K\alpha}$ radiation, final R indices ($I > 2\sigma(I)$): $R1 = 0.0614$, $wR2 = 0.1552$, $a = 11.0171(16)$, $b = 5.0148(8)$, $c = 21.889(3)$ Å, $\beta = 93.672(3)°$, $V = 1206.9(3)$ Å^3, $Z = 4$, number of reflections (measured/unique): 7157/4737 ($R_{int} = 0.0762$), number of observations: 2877 ($I > 2\sigma(I)$), 273 parameters; see reference [10].

[13] Crystal data for *trans*-**5l**: $C_{12}H_{13}Br_5O_2S$, $M_r = 620.83$, monoclinic, $P2(1)/n$, $Mo_{K\alpha}$ radiation, final R indices ($I > 2\sigma(I)$), $R_1 = 0.0763$, $wR2 = 0.1752$, $a = 15.211(7)$, $b = 8.598(4)$, $c = 15.846(7)$ Å, $\beta = 118.082(7)°$, $V = 1828.4(14)$ Å^3, $Z = 4$, number of reflections (measured/unique): 9294/3410 ($R_{int} = 0.1718$), number of observations: 1375 ($I > 2\sigma(I)$), 184 parameters; see reference [10].

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